Monday Lab

Date: - 2 September 2024

1. A. Write a program in C to input a number and check whether the number is prime number or not using for loop. How to check prime numbers using loop in C programming. Logic to check prime numbers in C programming.
   Example
   Input: Input any number: 17
   Output: 17 is prime number

   B. Write a C program to print Fibonacci series up to n terms using loop. How to generate Fibonacci series up to n terms using loops in C programming. Logic to print Fibonacci series in a given range in C programming.

   ## Fibonacci Formula:

   $F(n)=F(n-1)+F(n-2)$

   Where:

   - F(n) is the nth Fibonacci number.
   - F(n−1) is the previous Fibonacci number.
   - F(n−2) is the Fibonacci number before the previous one.

   ## Initial Conditions:

   - F(0)=0
   - F(1)=1

   Example
   Input Input number of terms: 10
   Output Fibonacci series: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34

2. Print a right-angled triangle of Fibonacci numbers with `n` rows. Each row `i` should contain the first `i` Fibonacci numbers. Use a nested loop to generate and print the numbers.

   Example:

   Input: `Number of rows: 4`

   Expected Output:

```
0
0 1
0 1 1
0 1 1 2
```

3. **Problem Description:** In the library, there is a penalty imposed for each late book return. If a book is returned late, the fine varies as follows: for the first 5 days, the penalty is 50 paise/day; for days 6 to 10, it's one rupee/day, and for delays exceeding 10 days, the fine is 5 rupees/day. Additionally, if a book is returned after 30 days, the member's library membership will be revoked.

   Write a C program that takes input for the number of days a member is overdue in returning a book and then displays the corresponding fine or an appropriate message.
   Example:
   Input: No of days the member is overdue – 7
   Output: The penalty charged is 4.50 Rs.
4. Problem Description:Write a C program that allows the user to calculate the area and perimeter (or circumference) of different geometric shapes: rect- angles, circles, and triangles. The program should present a menu with options for each shape and perform the calculations based on the user's choice.

   Explanation:

   - **Rectangle:**
     - Area: Area=length×width
     - Perimeter: Perimeter=2×(length+width)
   - **Circle:**
     - Area: Area=π×r2Area=π×r2
     - Circumference: Circumference=2×π×r
   - **Triangle:**
     - Area (using base and height): Area=1/2×base×height
     - Perimeter: Sum of all sides

   Example:
   Input: Geometry Calculator
   1. Calculate the area and perimeter of a rect- angle
   2. Calculate the area and circumference of a circle
   3. Calculate the area of a triangle

   Enter your choice (1/2/3) - 1
   Enter the length of the rectangle - 5.0
   Enter the width of the rectangle - 3.0
   Output: Area of the rectangle: 15.00 square units

Perimeter of the rectangle: 16.00 units

Input: Enter your choice (1/2/3) - 2
Enter the radius of the circle - 4.5
output: Area of the circle: 63.62 square units
Circumference of the circle: 28.27 units

Input: Enter your choice (1/2/3) - 3 Enter the base of the triangle - 3.0 Enter the height of the triangle - 4.0
Enter 3 sides of the triangle - 3 4 5
output: Area of the triangle: 6.00 square units
Perimeter of the triangle: 12 units